02 - Bayesian Decision Making

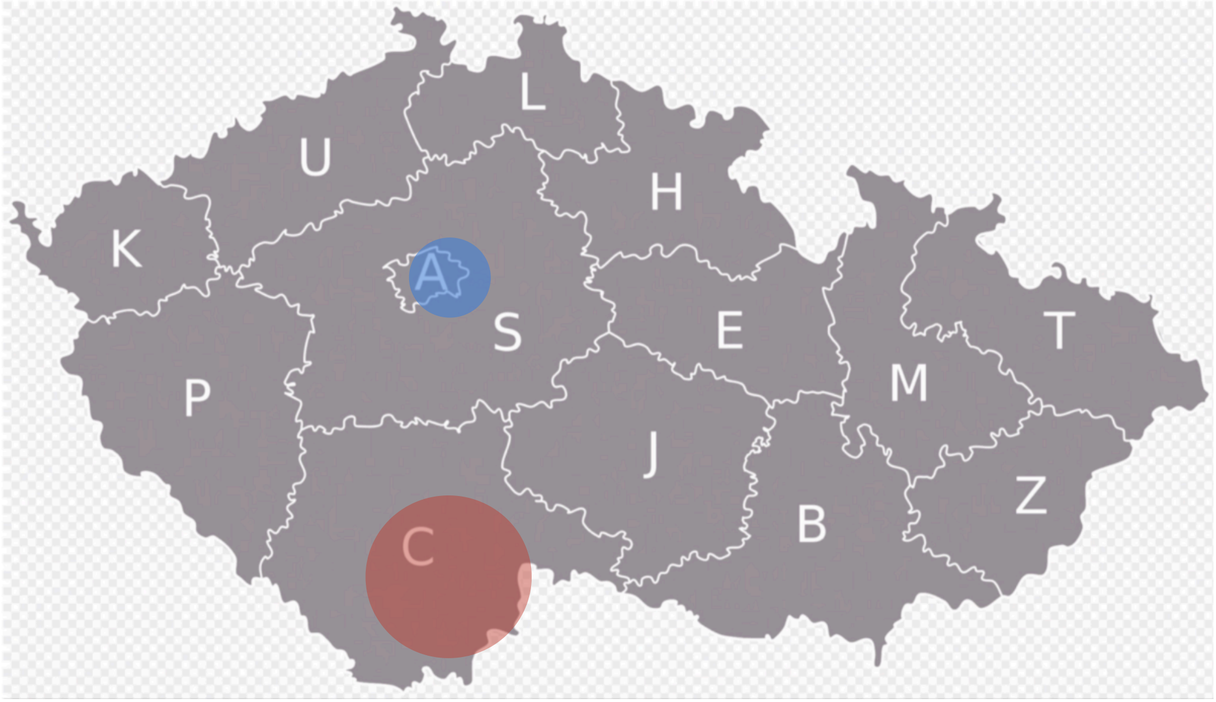
Pattern Recognition and Machine Learning, Lab 2

Michal Neoral
neoramic@fel.cvut.cz

2 October 2026

Motivation: licence plates

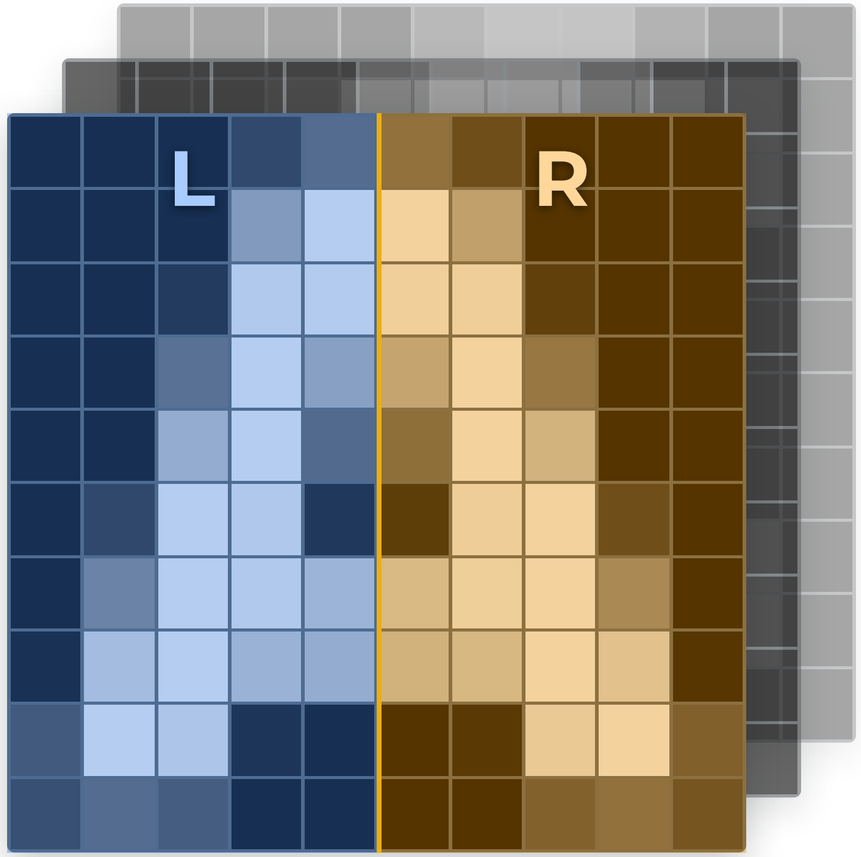
Kód	Kraj	2001–2004	Od 2004 do současnosti	Zatím nejvyšší řada [9]
A	Praha	1A7 4551	2AD 3824	8AJ
B	Jihomoravský kraj		6B2 6635	2BN
C	Jihočeský kraj	1C2 6377	7C2 5025	9C2
E	Pardubický kraj		2E7 7010	6E7
H	Královéhradecký kraj	2H2 7149	2H4 2270	7H8
J	Kraj Vysočina	2J2 6382	3J1 2272	6J9
K	Karlovarský kraj		3K3 2886	4K8
L	Liberecký kraj		2LO 2404	6L1
M	Olomoucký kraj	1M4 0000	6M1 3246	7M3
P	Plzeňský kraj	1P5 0858	7P2 4159	8P4
S	Středočeský kraj	3S9 6552	7S3 3237	5SK
T	Moravskoslezský kraj		6T9 9994	1TZ
U	Ústecký kraj		6U3 2183	1UH
Z	Zlínský kraj		4ZO 7472	6Z9



The second letter of an older plate tells the **region** (kraj).

! Our simplification: recognize only the letters **A** (Praha) and **C** (Jihočeský kraj).

The data and the features



N images of 10 × 10 pixels

How do we get a (discrete) feature x from an image of a letter?

L = sum of the pixel values in the **left** half

R = sum of the pixel values in the **right** half

$$x = g(L - R) \in X = \{-10, \ldots, 10\}$$

g quantizes the difference to a small set of integers

The Bayesian risk

❓ How do we define the Bayesian risk $R(q)$ of a strategy q?

$$R(q) = \sum_{x \in X} \sum_{k \in K} p_{XK}(x, k)\ W(k, q(x))$$

Symbol	Meaning	Licence plates	Mushrooms
X	observations $x \in X$: what we **know**	$x = g(L - R)$	*cap colour, smell, ring*
K	hidden states $k \in K$: what we **don't know**	letters $\{A, C\}$	*edible, poisonous*
D	decisions $d \in D$	$D = K = \{A, C\}$	*eat, leave it ($D \neq K$)*
p_{XK}	joint probability $p_{XK}(x, k)$	feature x with letter k	*a red smelly one is poisonous*
W	loss $W(k, d)$: price of d when the truth is k	0/1 loss	*poisonous eaten: hospital*
q	strategy $q : X \to D$	A or C for every x	*"red with dots: leave it"*

❓ What is the Bayesian (optimal) strategy q^*?

What do we know?

Known

- $p(x \mid k)$: likelihood
- $p(k)$: prior

Unknown

- $p_{XK}(x, k)$: joint probability
- it appears in the risk $R(q)$

How do we get $p_{XK}(x, k)$?

Derivation: the joint probability

Product rule $p_{XK}(x, k) = p(x \mid k)\, p(k) = p(k \mid x)\, p(x)$

Marginal $p(x) = \sum_{k \in K} p_{XK}(x, k) = \sum_{k \in K} p(x \mid k)\, p(k)$

Bayes' rule $p(k \mid x) = \dfrac{p(x \mid k)\, p(k)}{p(x)}$

The risk now contains only known quantities:

$$R(q) = \sum_{x \in X} \sum_{k \in K} p(x \mid k)\, p(k)\, W(k, q(x))$$

Discrete example

Bayes example: discrete x

$x = g(L - R) \in X = \{x_1, x_2, x_3, x_4\}, \quad K = D = \{A, C\}$

Priors: $p(A) = p(C) = 0.5$

Likelihoods $p(x \mid k)$

	x_1	x_2	x_3	x_4
A	0.3	0.2	0.1	0.4
C	0.7	0.2	0.1	0

Loss $W(k, d)$

$k \backslash d$	A	C
A	0	1
C	1	0

Strategy $q(x)$

x_1	x_2	x_3	x_4
A	A	C	C

? 1. What is the risk $R(q)$ of this strategy?

? 2. How do we find the optimal strategy q^*?

? 3. How do we determine the classification error?

Derivation: risk of the given strategy

$$R(q) = \sum_{x \in X} \sum_{k \in K} p(x \mid k)\, p(k)\, W(k, q(x))$$

With the 0/1 loss, only the wrong classes $k \neq q(x)$ contribute:

$$R(q) = \sum_{x \in X} \sum_{k \neq q(x)} p(x \mid k)\, p(k)$$

	x_1	x_2	x_3	x_4
decision $q(x)$	A	A	C	C
wrong class k	C	C	A	A
$p(x \mid k)\, p(k)$	$0.7 \cdot 0.5 = 0.35$	$0.2 \cdot 0.5 = 0.1$	$0.1 \cdot 0.5 = 0.05$	$0.4 \cdot 0.5 = 0.2$

$$R(q) = 0.35 + 0.1 + 0.05 + 0.2 = \mathbf{0.7}$$

Derivation: the optimal strategy

$$R(q) = \sum_{x \in X} \underbrace{\sum_{k \in K} p(x \mid k)\, p(k)\, W(k, q(x))}_{R(x,\, q(x)):\ \text{partial risk}}$$

Each term depends only on the single decision $q(x)$, so we minimize every term separately:

$$\min_q R(q) = \sum_{x \in X} \min_{d \in D} R(x, d), \qquad q^*(x) = \arg\min_{d \in D} R(x, d)$$

For the 0/1 loss and two classes:

$$R(x, A) = p(x \mid C)\, p(C), \qquad R(x, C) = p(x \mid A)\, p(A)$$

Example for x_1:

$$R(x_1, A) = 0.7 \cdot 0.5 = 0.35, \qquad R(x_1, C) = 0.3 \cdot 0.5 = \mathbf{0.15}, \qquad q^*(x_1) = C$$

Discrete example: results

1. Risk of the given strategy

$$R(q) = p(A)\,(0.1 + 0.4) + p(C)\,(0.7 + 0.2) = \mathbf{0.7}$$

3. Classification error

$$\text{error} = \frac{\#\,\text{misclassified}}{\#\,\text{all} = N}$$

2. Partial risk and the optimal strategy

$$R(x, d) = \sum_{k \in K} p(x \mid k)\, p(k)\, W(k, d), \qquad q^*(x) = \arg\min_{d \in D} R(x, d)$$

	x_1	x_2	x_3	x_4
$R(x, A)$	0.35	**0.1**	**0.05**	**0**
$R(x, C)$	**0.15**	**0.1**	**0.05**	0.2
$q^*(x)$	**C**	**A** or **C**	**A** or **C**	**A**

$R(q^*) = 0.15 + 0.1 + 0.05 + 0 = \mathbf{0.3}$. For x_2, x_3 both decisions are optimal (a tie), a preference must be agreed on.

Continuous x

Bayes example: continuous x

$x = L - R \in X = \mathbb{R}$ is now **continuous**

Likelihood: normal distribution

$$p(x \mid k) = \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2} \right)$$

Given:

μ_A, μ_C, σ_A, σ_C, $p(A)$, $p(C)$, W = 0/1 loss

? How does the Bayesian risk $R(q)$ change for a continuous x?

? What is the optimal strategy $q^*(x)$?

Derivation: optimal strategy, continuous x

$$R(q) = \int_X \sum_{k \in K} p_{XK}(x, k)\, W(k, q(x))\, dx = \int_X p(x) \sum_{k \in K} p(k \mid x)\, W(k, q(x))\, dx$$

$p(x) \geq 0$, so we minimize the integrand separately for every x:

$$q^*(x) = \arg\min_{d \in D} \sum_{k \in K} p(k \mid x)\, W(k, d)$$

For the 0/1 loss: $\sum_{k \neq d} p(k \mid x) = 1 - p(d \mid x)$, therefore

$$q^*(x) = \arg\max_{d \in D} p(d \mid x) = \arg\max_{d \in D} \frac{p(x \mid d)\, p(d)}{p(x)} = \arg\max_{d \in D} p(x \mid d)\, p(d)$$

Two classes: a quadratic inequality

We decide A where $p(x \mid A)\, p(A) \geq p(x \mid C)\, p(C)$:

$$p(A)\, \frac{1}{\sigma_A \sqrt{2\pi}}\, e^{-\frac{(x-\mu_A)^2}{2{\sigma_A}^2}} \;\geq\; p(C)\, \frac{1}{\sigma_C \sqrt{2\pi}}\, e^{-\frac{(x-\mu_C)^2}{2{\sigma_C}^2}}$$

$\sqrt{2\pi}$ cancels, both sides are positive, so we take the logarithm:

$$\ln p(A) - \ln \sigma_A - \frac{(x-\mu_A)^2}{2{\sigma_A}^2} \;\geq\; \ln p(C) - \ln \sigma_C - \frac{(x-\mu_C)^2}{2{\sigma_C}^2}$$

Move the terms without x to the right:

$$-\frac{(x-\mu_A)^2}{2{\sigma_A}^2} + \frac{(x-\mu_C)^2}{2{\sigma_C}^2} \;\geq\; \ln\left(\frac{\sigma_A}{\sigma_C} \cdot \frac{p(C)}{p(A)}\right) =: L$$

Two classes: collecting the powers of x

$$-\frac{(x-\mu_A)^2}{2{\sigma_A}^2}+\frac{(x-\mu_C)^2}{2{\sigma_C}^2} \geq L \quad \Big| \cdot 2{\sigma_A}^2{\sigma_C}^2 > 0$$

$$-{\sigma_C}^2(x-\mu_A)^2+{\sigma_A}^2(x-\mu_C)^2 \geq 2{\sigma_A}^2{\sigma_C}^2 L$$

Expand the squares:

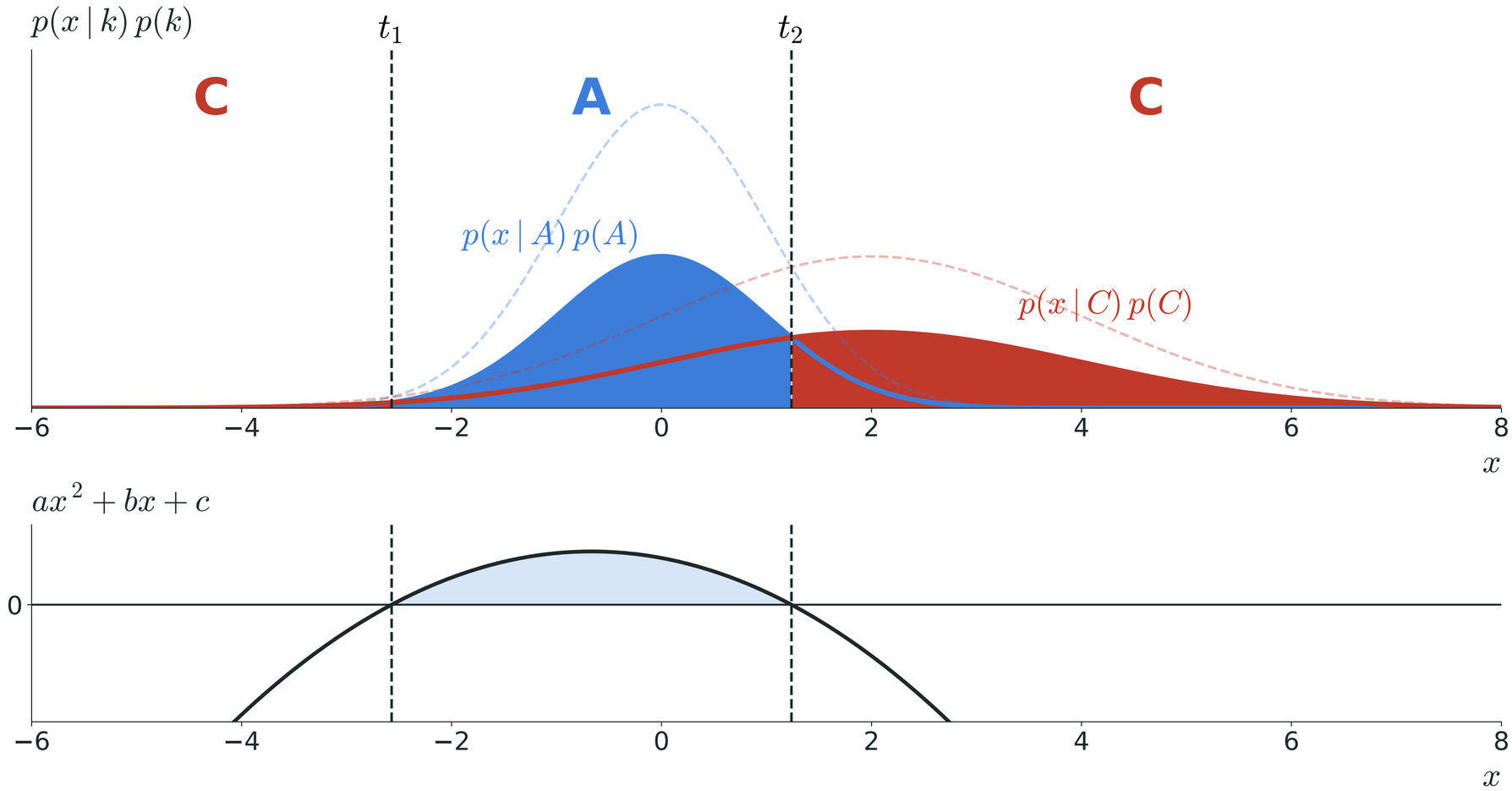
$$-{\sigma_C}^2x^2+2{\sigma_C}^2\mu_A x-{\sigma_C}^2{\mu_A}^2+{\sigma_A}^2x^2-2{\sigma_A}^2\mu_C x+{\sigma_A}^2{\mu_C}^2-2{\sigma_A}^2{\sigma_C}^2 L \geq 0$$

Group by the powers of x:

$$\left({\sigma_A}^2-{\sigma_C}^2\right)x^2+2\left({\sigma_C}^2\mu_A-{\sigma_A}^2\mu_C\right)x+{\sigma_A}^2{\mu_C}^2-{\sigma_C}^2{\mu_A}^2-2{\sigma_A}^2{\sigma_C}^2 L \geq 0$$

$$a={\sigma_A}^2-{\sigma_C}^2 \qquad b=2\left({\sigma_C}^2\mu_A-{\sigma_A}^2\mu_C\right) \qquad c={\sigma_A}^2{\mu_C}^2-{\sigma_C}^2{\mu_A}^2-2{\sigma_A}^2{\sigma_C}^2 L$$

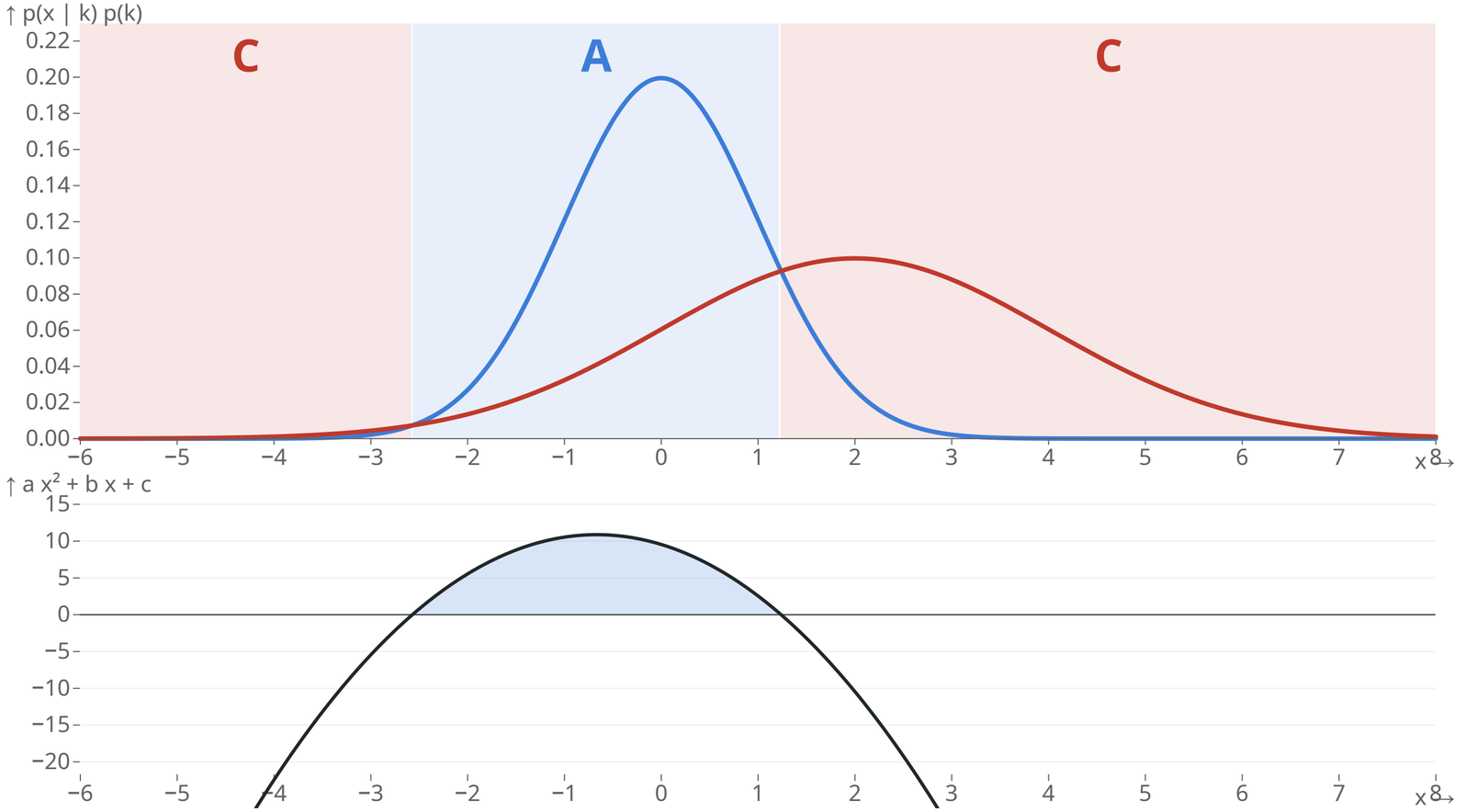
Decision regions



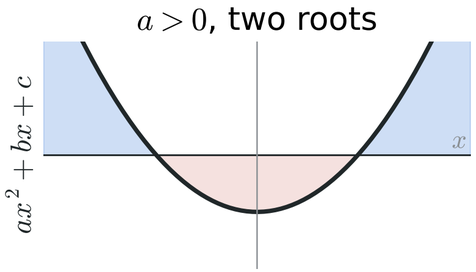
The thresholds are the roots of $ax^2 + bx + c = 0$, class A where it is ≥ 0

Decision regions: try it

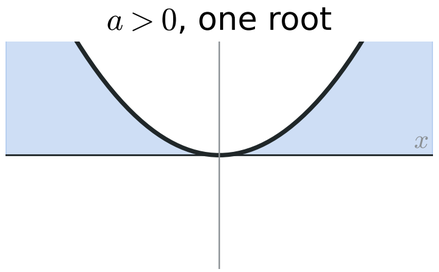
μA	0
σA	1
μC	2
σC	2
p(A)	0.5



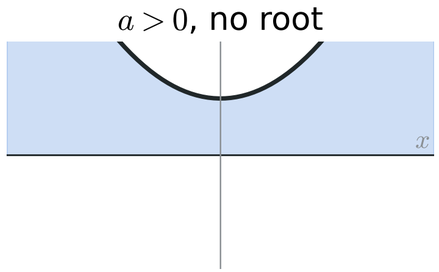
Watch out: all the cases of $ax^2 + bx + c \geq 0$



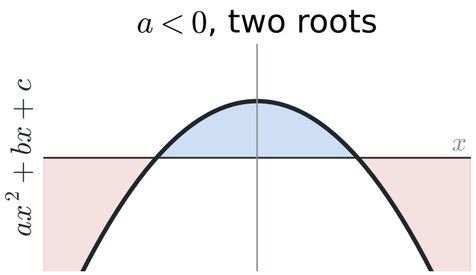
A | C | A



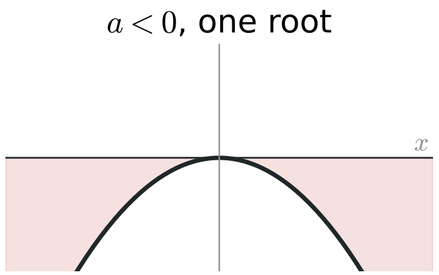
A everywhere



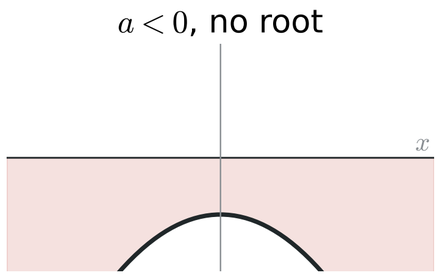
A everywhere



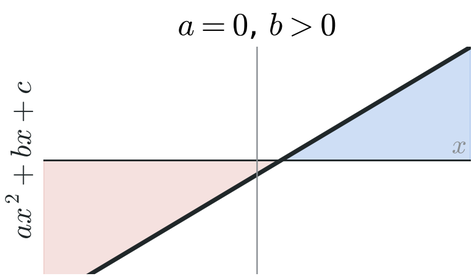
C | A | C



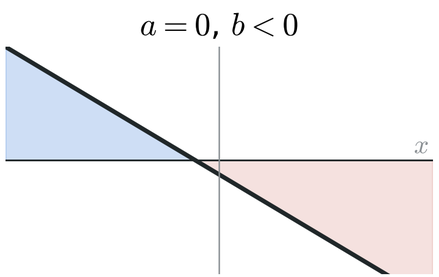
C (A in one point)



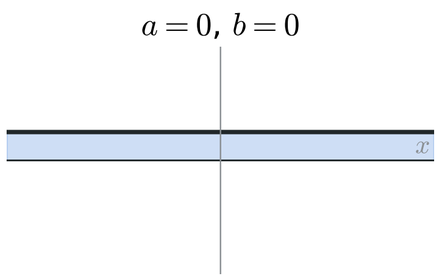
C everywhere



C | A



A | C



A if $c \geq 0$, else C

Which intervals belong to class A in each case?

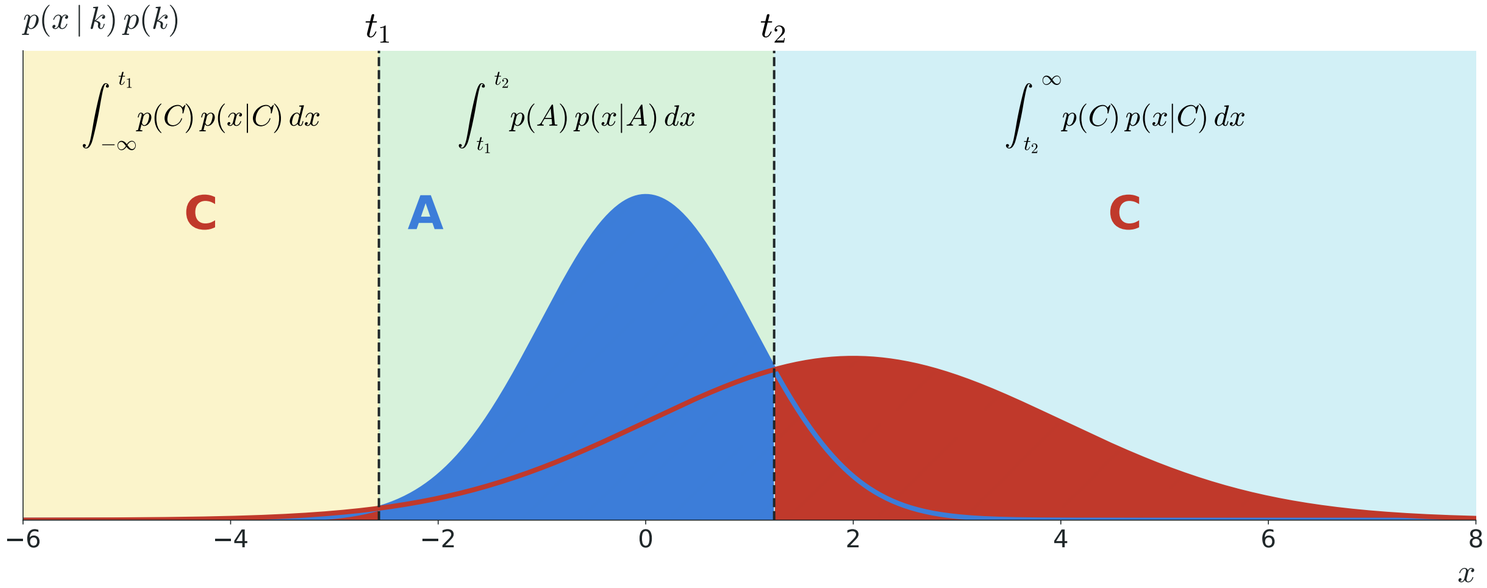
Handle **every case** in your code, including $a = 0$ and no real roots.

Output of the assignment



The strategy is stored as two thresholds `q['t1'] ⩽ q['t2']` and the decisions for the three intervals, e.g. `q['decision'] = ['C', 'A', 'C']`.

The risk for continuous x



Compute the risk $R(q)$ of your strategy.

Hint: the area under a normal density up to t is `scipy.stats.norm.cdf(t, mu, sigma)`

$$R(q) = 1 - \left(\int_{-\infty}^{t_1} p(C)\, p(x|C)\, dx + \int_{t_1}^{t_2} p(A)\, p(x|A)\, dx + \int_{t_2}^{\infty} p(C)\, p(x|C)\, dx \right)$$

Derivation: the risk for continuous x

$$R(q) = \int_{-\infty}^{\infty} \sum_{k \in K} p(k)\, p(x \mid k)\, W(k, q(x))\, dx$$

0/1 loss: all classes except the decided one $q(x)$:

$$R(q) = \int_{-\infty}^{\infty} \Big(\sum_{k \in K} p(k)\, p(x \mid k) \;-\; p(q(x))\, p(x \mid q(x)) \Big)\, dx$$

The first part integrates to one:

$$\int_{-\infty}^{\infty} \sum_{k} p(k)\, p(x \mid k)\, dx = \sum_{k} p(k) \int_{-\infty}^{\infty} p(x \mid k)\, dx = \sum_{k} p(k) = 1$$

$$R(q) = 1 - \int_{-\infty}^{\infty} p(q(x))\, p(x \mid q(x))\, dx$$

The integral is the probability of a correct decision.

Derivation: splitting into intervals

q is constant on the intervals (t_{i-1}, t_i) with decisions d_1, d_2, d_3, where $t_0 = -\infty$, $t_3 = \infty$:

$$R(q) = 1 - \sum_{i=1}^{3} p(d_i) \int_{t_{i-1}}^{t_i} p(x \mid d_i)\, dx$$

Cumulative distribution function of class k, in code `norm.cdf(t, mu, sigma)`:

$$F_k(t) = \int_{-\infty}^{t} p(x \mid k)\, dx, \qquad \int_a^b p(x \mid k)\, dx = F_k(b) - F_k(a)$$

with $F_k(-\infty) = 0$ and $F_k(\infty) = 1$.

For the decisions $C \mid A \mid C$:

$$R(q) = 1 - \Big(p(C)\, F_C(t_1) + p(A)\big(F_A(t_2) - F_A(t_1)\big) + p(C)\big(1 - F_C(t_2)\big) \Big)$$

Derivation: a numerical example

Parameters:

$\mu_A = 0,\ \sigma_A = 1,\ \mu_C = 2,\ \sigma_C = 2,\ p(A) = 0.5$

Thresholds:

$t_1 = -2.571,\ t_2 = 1.238$, decisions $C \mid A \mid C$

$$F_C(t_1) = 0.0111, \quad F_A(t_1) = 0.0051, \quad F_A(t_2) = 0.8921, \quad F_C(t_2) = 0.3515$$

$$R(q) = 1 - 0.5\left(0.0111 + (0.8921 - 0.0051) + (1 - 0.3515)\right) = 1 - 0.7733 = \mathbf{0.2267}$$

Check: the areas of the wrong class give the same result:

$$R(q) = p(A)\left(F_A(t_1) + 1 - F_A(t_2)\right) + p(C)\left(F_C(t_2) - F_C(t_1)\right) = 0.2267$$

Take-away for the lab

- ☑ **Discrete x:** risk of a given strategy, the optimal strategy, classification error
- ☑ **Continuous x:** coefficients a, b, c of the quadratic inequality
- ☑ Thresholds `t1`, `t2` and the **decisions** for the three intervals
- ☑ **Risk** of the strategy using `scipy.stats.norm.cdf`
- ☑ Watch out for **all the cases** ($a = 0$, one or no real root)


"That's all Folks!"